Unsymmetric spectral elements under non-polynomial displacement fields: distortion sensitivity and convergence for the thick cylinder

USEM-MATLAB study `studies/thick_cylinder`

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Abstract

Two unsymmetric (parametric test / metric trial) spectral elements of the USEM code base are studied for the thick cylinder of Rajendran [5, Sec. 5.9] with $p = 2, \ldots, 8$ on gmsh meshes whose distortion is kept fixed under refinement ($\alpha = 0°$–$90°$, including folded elements), complemented by manufactured polynomial and analytic non-polynomial solutions on the same meshes: PM (metric trial space Q_p in skew coordinates) and PH (hybrid trial space: complete polynomials P_p in Cartesian coordinates, complemented by parametric functions); the conventional spectral element PP serves as reference (1512 solutions, verified against the code base). *Findings.* (i) Both unsymmetric elements reproduce a degree-5 polynomial field for $p \geq 5$; for PM the round-off level grows with distortion and p (up to 10^{-4} at $\alpha = 45°$), for PH it stays below 10^{-9} on all meshes on which the element can be built. (ii) For non-polynomial fields distortion immunity is lost by both. PM remains distortion tolerant and optimally convergent only for moderate distortion ($\alpha = 15°$, $p \leq 6$; $\alpha = 30°$, $p \leq 3$); beyond, its convergence becomes erratic. PH converges with the optimal order p for all p on all non-folded meshes, is quasi-optimal (error 0.78–1.00 times its interpolation error) and up to $2.6 \cdot 10^4$ times more accurate than PP, which drops to order $\approx \lfloor p/2 \rfloor$. (iii) The loss of PM is caused by its metric interpolation (Lebesgue constants up to 10^8): the largest singular values of the stiffness matrix grow up to 340-fold relative to PP, and single low singular modes carry 21–68 % of the error; PH keeps Lebesgue constants and spectra close to those of PP. (iv) PH does not remove all limits: on folded meshes its error is still large for $p \leq 3$ and its interpolation matrix becomes singular for some folded elements at $p \geq 7$; for the Lamé field at $\alpha = 15°$, $p \leq 7$, it is up to 4 times less accurate than PM.

1 Objective

Unsymmetric (Petrov–Galerkin) elements with parametric test and metric trial functions reproduce every displacement field of total degree $\leq p$ exactly on arbitrarily distorted meshes, including meshes with $\det \boldsymbol{J} \leq 0$ (*distortion immunity*) [5, 6]. Displacement fields of practical problems are, however, not polynomial. Rajendran [5] observed for the thick cylinder (Section 5.9 of that paper, 8-node element) that the energy-norm convergence rate decreases from about 2 to 1.85, 1.42 and 1.21 as the distortion angle increases from 0° to 30°, 45° and 60°, and concluded that conditions C1–C3 do not guarantee optimal convergence for non-polynomial fields. This study revisits the question for the unsymmetric spectral elements of the code base, $p = 2, \ldots, 8$:

1. Is distortion resistance lost for non-polynomial fields?
2. If so, are optimal convergence rates nevertheless retained?
3. If both are lost, what is the practical benefit of unsymmetric elements?
4. Does a hybrid trial space, which uses metric (Cartesian) polynomials only for the complete space $P_p(\boldsymbol{x})$ and parametric functions for the remainder (Section 2.1), remove the deficiencies while keeping the polynomial exactness?

The thick cylinder is complemented by manufactured solutions on the same geometry and meshes: a polynomial field (degree 5), for which the element is immune for $p \geq 5$, and an analytic non-polynomial field. All statements below are based on the computed data in `results/` (reproducible with the scripts listed in Section 7); the conventional spectral element on the same meshes serves as reference. Questions 1–3 concern the unsymmetric element PM in its original form, question 4 the hybrid element PH.

2 Formulation and set-up

2.1 Elements

The three elements use the same geometry, the same nodes and the same test functions: Lagrange polynomials $N_i(\boldsymbol{\xi}) \in Q_p$ on the $(p+1)^2$ Gauss–Lobatto–Legendre (GLL) nodes of the reference square, mapped isoparametrically, $\boldsymbol{x}(\boldsymbol{\xi}) = \sum_i N_i(\boldsymbol{\xi})\, \boldsymbol{x}_i$. They differ in the trial functions.

- PM (unsymmetric spectral element of the code base, `Ke_2D.m`): the trial functions are *metric* functions $M_i^e(\boldsymbol{x}) \in Q_p(\hat{x}, \hat{y})$, defined element by element through $M_i^e(\boldsymbol{x}_j) = \delta_{ij}$ in the corotational skew coordinates $(\hat{x}, \hat{y})^{\mathsf{T}} = \boldsymbol{J}_0^{-\mathsf{T}}(\boldsymbol{x} - \boldsymbol{x}_0)$ ($\boldsymbol{J}_0$, $\boldsymbol{x}_0$: Jacobian and image of the element centre) [7]. As $(\hat{x}, \hat{y})$ is affine in $\boldsymbol{x}$, the span of $\{M_i^e\}$ contains $P_p(\boldsymbol{x})$ (all polynomials of total degree $\le p$ in Cartesian coordinates) for *every* element shape. With $\boldsymbol{B}_L^* = \boldsymbol{B}_L \det \boldsymbol{J}$ (built from $\operatorname{adj} \boldsymbol{J}\, \nabla_{\boldsymbol{\xi}} N_i$),

$$\boldsymbol{K}^e = \int_{-1}^{1}\int_{-1}^{1} (\boldsymbol{B}_L^*)^{\mathsf{T}} \boldsymbol{D}\, \boldsymbol{B}_R \, \mathrm{d}\xi \, \mathrm{d}\eta, \tag{1}$$

  i.e. the Jacobian determinant cancels and the integrand is a polynomial in $\boldsymbol{\xi}$; this is the Jacobian-unlimited formulation PM-QUAD8* of Rajendran [5] generalised to degree p (conditions C1–C3 of that paper). The trial space is non-conforming (the M_i^e are not continuous across element edges). The code base solves for the M_i^e in the Lagrange basis on the GLL points of the bounding box of the skew node coordinates (`p.metricBasis = 'lagrange'`, default); this changes the conditioning $\kappa(\boldsymbol{V})$ of the interpolation matrix, not the trial space, so all errors are those of the monomial basis up to round-off.

- PH (unsymmetric spectral element with hybrid trial space, `p.metricMethod = 2`): same test functions, Jacobian cancellation and (1), but the trial functions span

$$T_p^e = P_p(\hat{x}, \hat{y}) \oplus \operatorname{span}\{L_a(\xi) L_b(\eta) :\ a, b \le p,\ a + b > p\} \tag{2}$$

  (L_a: Legendre polynomials), i.e. metric polynomials only for the complete space $P_p(\boldsymbol{x})$, which guarantees completeness and polynomial exactness, and parametric functions for the remaining $(p+1)^2 - \frac{1}{2}(p+1)(p+2)$ functions. $\dim T_p^e = (p+1)^2$; on bilinear elements $T_p^e = Q_p(\boldsymbol{\xi})$, and on affine elements $T_p^e = Q_p(\boldsymbol{x})$ (`docs`, chapter on the metric basis). The strains of the complement need $\boldsymbol{J}^{-1}$, so PH is undefined where $\det \boldsymbol{J} = 0$ at a quadrature point (not encountered here).

- PP (conventional spectral element, reference): parametric functions as test and trial functions, $\boldsymbol{K}^e = \int \boldsymbol{B}_L^{\mathsf{T}} \boldsymbol{D} \boldsymbol{B}_L \det \boldsymbol{J} \, \mathrm{d}\xi \, \mathrm{d}\eta$ (`tests/helpers/usemKeSymmetric.m`).

The load vectors are identical for the three elements (parametric test functions), $\boldsymbol{f}^e = \int N_i\, \boldsymbol{b} \det \boldsymbol{J} \, \mathrm{d}\xi \, \mathrm{d}\eta + \int_{\Gamma_N} N_i\, \bar{\boldsymbol{t}} \, \mathrm{d}s$. The stiffness matrices use the default rule of `Ke_2D.m`, $2p \times 2p$ Gauss points; loads and error integrals use $(3p+2)^2$ points. The element matrices of the study (`cyl_Ke_PM.m`, `cyl_Ke_PP.m`) are vectorised versions of `Ke_2D.m` and `usemKeSymmetric.m`; they agree with them to round-off (Table 1). `ShapeFunctions2D_Metric` stops with an error when the interpolation matrix of an element is singular to working precision (rcond $< \varepsilon$, both unsymmetric elements); such cases are marked "rej.".

2.2 Model and mesh (gmsh)

Thick cylinder under internal pressure (Section 5.9 of [5]): quarter annulus $a = 1 \le r \le b = 3$, $E = 1000$, $\nu = 0.25$, plane stress, internal pressure $p_i = 1000$, symmetry conditions $u_x = 0$ on $x = 0$ and $u_y = 0$ on $y = 0$. The geometry (exact circular arcs, built-in kernel) and the structured mesh are generated with gmsh [3] (`mesh/gen_cylinder_meshes.py`): a transfinite mesh of $2N \times 2N$ eight-node quadrilaterals, i.e. $N \times N$ super-elements of 2×2 elements ($N = 1, 2, 4, 8$; element size $h \propto 1/N$). Mesh distortion follows mesh sequence B of [5, Fig. 19]: in every super-element the mid-side nodes of the four internal edges are rotated counter-clockwise about the centre node by the angle $\alpha \in \{0°, 15°, 30°, 45°, 60°, 90°\}$ (Fig. 1). The pattern is repeated on every super-element, so the *relative* distortion does not decrease under refinement: the element maps are not asymptotically affine. For $\alpha \ge 60°$ (and $\alpha = 45°$ for $N \ge 2$), $\det \boldsymbol{J}$ changes sign inside elements (folded elements, Table 2). Each mesh is written as an ABAQUS input file (corner nodes first, node and element sets, pressure `*Dsload`) that also runs with `code/Start_Sim.m`.

The 8-node mesh is lifted to degree p by `pSpectralElement_NodesTopology.m`. Element edges on $r = a$ and $r = b$ are then replaced by the exact arcs (Gordon–Hall blending [4]); interior edges remain the parabolas of the 8-node mesh. The degree-p geometry thus interpolates the circle at the GLL points; the signed area converges to 2π spectrally ($|A_h - 2\pi| < 10^{-14}$ for $p = 8$) and is independent of α because folded parts cancel (Table 1).

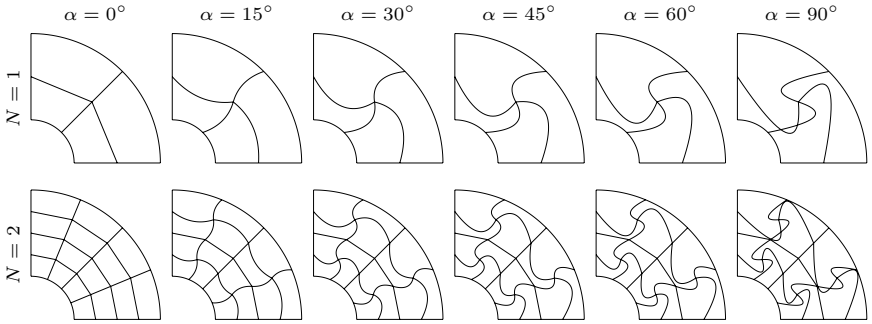


Figure 1: Meshes generated with gmsh for $N = 1$ (2×2 elements) and $N = 2$ (4×4 elements); the internal mid-side nodes of every super-element are rotated by α. Meshes for $N = 4, 8$ repeat the pattern on N^2 super-elements.

2.3 Problems

1. *Lamé* (non-polynomial, rational): $u_r = \frac{a^2 p_i\, r}{E(b^2-a^2)}\big[(1-\nu) + (1+\nu)\frac{b^2}{r^2}\big]$, $u_\theta = 0$ [5, Eq. (25)]; symmetry conditions on the straight edges, tractions on the arcs.

2. *Manufactured polynomial* field of total degree 5 ($X = x/3$, $Y = y/3$):
   $u_x = \frac{1}{10}(1 + X + 2Y + XY - 3X^2Y + X^3 + 2X^4Y - X^2Y^3 + X^5 - 2Y^5)$,
   $u_y = \frac{1}{10}(1 - X + Y^2 + 2X^2Y - X^3 + XY^3 - 3X^3Y^2 + Y^5 + 2XY^4 - X^5)$.

3. *Manufactured smooth non-polynomial* (entire) field: $u_x = \frac{1}{10}\big(\sin 3X \cos 2Y + e^{XY}\big)$, $u_y = \frac{1}{10}\big(\cos 2X \sin 3Y + Y e^{-X}\big)$.

For the manufactured solutions (MMS) both displacement components are prescribed on $x = 0$ and $y = 0$ (nodal values) and the tractions $\boldsymbol{\sigma}(\boldsymbol{u})\boldsymbol{n}_h$ on the arcs. Strains, stresses and the body force $\boldsymbol{b} = -\operatorname{div}\boldsymbol{\sigma}(\boldsymbol{u})$ are derived symbolically with the operators of the MMS module of the code base (`MMS_2D/differentialOperators`, as in `examples_Elasticity.m`); for the Lamé field this yields $\boldsymbol{b} \equiv \boldsymbol{0}$. On the arcs $\boldsymbol{n}_h$ is the normal of the discrete boundary, so that the discrete problem on Ω_h has the exact field as solution whenever it lies in the trial space (with the exact normal, the polynomial patch test fails at the level of the geometry error, e.g. $e_E \approx 3 \cdot 10^{-6}$ for $p = 5$, $N = 1$).

2.4 Error measures

With the strain field of the respective trial functions (metric for PM, hybrid for PH, parametric for PP),

$$e_E^2 = \frac{\sum_e \int_{\hat\Omega} (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_h^e)^{\mathsf T} \boldsymbol{D} (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_h^e) \det \boldsymbol{J}\, \mathrm{d}\xi\, \mathrm{d}\eta}{\int_\Omega \boldsymbol{\varepsilon}^{\mathsf T} \boldsymbol{D} \boldsymbol{\varepsilon}\, \mathrm{d}\Omega}, \qquad e_{L^2}^2 = \frac{\sum_e \int_{\hat\Omega} |\boldsymbol{u} - \boldsymbol{u}_h^e|^2 \det \boldsymbol{J}\, \mathrm{d}\xi\, \mathrm{d}\eta}{\int_\Omega |\boldsymbol{u}|^2\, \mathrm{d}\Omega}, \tag{3}$$

and the nodal error $e_n = \max_i |\boldsymbol{u}_h(\boldsymbol{x}_i) - \boldsymbol{u}(\boldsymbol{x}_i)| / \max_i |\boldsymbol{u}(\boldsymbol{x}_i)|$. The signed $\det \boldsymbol{J}$ is used, consistent with (1): on folded elements each element contributes with the degree of its map. For PP on folded meshes the parametric strain is singular where $\det \boldsymbol{J} = 0$, so e_E is not meaningful there; e_n is defined for all cases. Observed rates are $r = \log_2(e_N / e_{2N})$. For smooth solutions and shape-regular, asymptotically affine meshes, optimal rates are p (e_E) and $p + 1$ (e_{L^2}) [2].

3 Verification

Table 1 summarises the cross-checks of the study routines against the code base (`matlab/cyl_verify.m`, full output in `results/verification.txt`). The study was run with GNU Octave 8.4 (MATLAB is not available in the build environment); three Octave stand-ins (`matlab/octave_compat`) replace MATLAB functions used by the code base (`contains`, `decomposition`, the third output of `unique(...,'stable')`). Compared with the first run of this study (monomial basis of the metric functions), the PP results are bitwise identical. The PM trial space is the same, so only round-off differs: on the meshes without folds, PM errors more than ten times above the round-off level (Section 4.2) change by at most 7 %; errors at the round-off level and errors on folded meshes, where the PM solution is dominated by the amplified round-off, change in either direction (smaller in 235, larger in 127 of the 362 values that changed by more than 0.1 %, by up to a factor 10.5). 7 PM cases per problem (folded meshes, $p \geq 6$) are now rejected. Table 2 lists the number of folded elements.

Table 1: Verification against the code base (relative differences).

check	result
element stiffness: `cyl_Ke_PM` vs `Ke_2D` (`p.metricMethod` = 1, 2), `cyl_Ke_PP` vs `usemKeSymmetric` ($\alpha = 0°, 60°, 90°$; $p = 2, 5, 8$)	$\leq 5.0 \cdot 10^{-15}$ (PM), $\leq 2.4 \cdot 10^{-15}$ (PH), $\leq 5.7 \cdot 10^{-12}$ (PP, folded elements); PM and PH rejected by the code base for $\alpha = 90°$, $p = 8$
global stiffness vs `AssemblyFEM` ($N = 2$; $\alpha = 0°, 60°$; $p = 2, 4$)	$\leq 1.1 \cdot 10^{-15}$
pressure load of the ABAQUS file (`NeumannBCs/CreatePress`) vs $\int N \boldsymbol{\sigma} \boldsymbol{n}_h \, \mathrm{d}s$ (Lamé, $N = 1$, $p = 2, \ldots, 8$)	$2 \cdot 10^{-3}$ ($p = 2$) $\to 1.7 \cdot 10^{-11}$ ($p = 8$): geometry error of the arc; resultant $1000 = p_i a$ in both
static solution with the code-base pipeline (`AssemblyFEM`, `NeumannBCs`, BCs of the ABAQUS file) vs study routines	$\leq 4.2 \cdot 10^{-11}$ ($\alpha = 0°$), $\leq 3.0 \cdot 10^{-7}$ ($\alpha = 60°$, round-off amplified by the conditioning)
signed area -2π ($p = 8$, all meshes)	$\leq 1.9 \cdot 10^{-14}$
exact fields (Lamé): $\boldsymbol{b} = \boldsymbol{0}$, $\sigma_r(a) = -p_i$, $\sigma_\theta(a) = 1250$, $\sigma_r(b) = 0$, $u_r(a) = 1.5$	exact
polynomial patch test, $p \geq 5$ (Table 3): PM, $\alpha \leq 30°$; PH, all meshes (cases not rejected)	$e_n \leq 7.6 \cdot 10^{-9}$ (PM), $e_n \leq 9.9 \cdot 10^{-10}$ (PH)

Table 2: Folded elements ($\det \boldsymbol{J} \leq 0$ somewhere in the element) / number of elements.

	$N = 1$	$N = 2$	$N = 4$	$N = 8$
$\alpha = 0°$	0/4	0/16	0/64	0/256
$\alpha = 15°$	0/4	0/16	0/64	0/256
$\alpha = 30°$	0/4	0/16	0/64	0/256
$\alpha = 45°$	0/4	2/16	8/64	32/256
$\alpha = 60°$	2/4	6/16	32/64	112/256
$\alpha = 90°$	4/4	16/16	64/64	256/256

4 Results

The complete data (6 angles × 4 meshes × 7 degrees × 3 elements × 3 problems = 1512 solutions) are in `results/raw/study_a*.csv`. Rates are given for the finest pair $N = 4 \to 8$; "–" marks errors at the round-off level (for PM and PH with $p \geq 5$: less than ten times the error of the polynomial field on the same mesh, which is the empirical round-off level, Section 4.2); "rej." marks cases in which the code base rejects the element (singular interpolation matrix, Section 2.1); † marks angles with folded elements on the meshes involved (Table 2).

4.1 Reproduction of Section 5.9 for $p = 2$

For $p = 2$ and the meshes $N = 1, 2, 4$ of the paper, the least-squares slopes of e_E (Lamé) are 1.90, 1.95, 1.14, 1.57 for PM, 1.81, 1.86, 1.85, 1.77 for PH and 1.68, 1.11, 1.07, 0.69 for PP at $\alpha = 0°, 30°, 45°, 60°$ (Rajendran [5, Fig. 20]: PM-QUAD8* 2.1, 1.85, 1.42, 1.21; QUAD8 1.9, 0.95, 0.94, 0.98). The trends of PM and PP agree with the paper: the conventional element drops to first order on distorted meshes, the unsymmetric element stays close to second order for $\alpha \leq 30°$ and becomes erratic beyond. Exact agreement is not expected, since the present $p = 2$ element is the 9-node Lagrange element with Q_2 metric functions in skew coordinates, not the 8-node serendipity element of the paper. PH keeps a slope of 1.77–1.86 up to $\alpha = 60°$. On the finer pair $N = 4 \to 8$, PM and PH ($p = 2$) converge with rate 1.94–1.97 for all $\alpha \leq 45°$, PP with 1.01–1.07 for $\alpha \geq 15°$ (Tables 5 and 7).

4.2 Polynomial field: distortion immunity

For $p \geq 5$ the degree-5 field lies in both unsymmetric trial spaces on every element. Table 3 shows the nodal error. PM: on the meshes without folds ($\alpha \leq 30°$) it is $\leq 7.6 \cdot 10^{-9}$ for all N, i.e. the field is reproduced up to round-off, as predicted; the round-off level, however, grows with α and p and reaches $1.1 \cdot 10^{-4}$ at $\alpha = 45°$, $N = 8$, $p = 8$, and $1.6 \cdot 10^{-2}$ on the folded mesh $\alpha = 90°$, $N = 4$, $p = 8$. PH: the error is $\leq 5.1 \cdot 10^{-13}$ for $\alpha \leq 30°$ and $\leq 9.9 \cdot 10^{-10}$ on all meshes, folded ones included, on which the element can be built, i.e. distortion immunity holds also in floating point. Both elements are rejected for the same seven cases ($\alpha = 60°, 90°$, $p \geq 6$, except PH for $\alpha = 60°$, $N = 8$, $p = 6$), in which the interpolation matrix is singular to working precision (rcond = 10^{-20}–10^{-17}). A dedicated test (`results/conditioning.txt`) integrates the stiffness exactly ($n_q \geq p^2 + p/2$ points, Section 4.7) and evaluates the residual $\boldsymbol{K}\boldsymbol{u}_I - \boldsymbol{F}$ of the exact nodal values, which vanishes in exact arithmetic. For PM it is $5 \cdot 10^{-14}$ at $\alpha = 0°$, $p = 5$, but $9.9 \cdot 10^{-7}$ ($e_n = 4 \cdot 10^{-5}$) at

Table 3: Polynomial field (degree 5): relative nodal error e_n for $p \geq 5$ (exact in exact arithmetic for PM and PH).

	$N = 1$				$N = 8$			
	$p = 5$	$p = 6$	$p = 7$	$p = 8$	$p = 5$	$p = 6$	$p = 7$	$p = 8$
PM (unsymmetric, skew coordinates)								
$\alpha = 0°$	2.2e-14	1.5e-13	1.5e-12	1.6e-11	1.3e-13	1.1e-13	2.8e-13	3.3e-13
$\alpha = 15°$	2.4e-11	1.3e-12	3.1e-12	3.8e-10	2.7e-13	2.6e-13	1.0e-11	1.8e-11
$\alpha = 30°$	7.2e-12	2.4e-12	1.1e-10	4.3e-10	3.2e-11	1.2e-11	1.4e-9	5.1e-9
$\alpha = 45°$	1.9e-11	2.2e-10	6.6e-9	2.7e-6	9.8e-10	7.9e-8	5.6e-5	1.1e-4
$\alpha = 60°$	1.6e-10	8.5e-10	3.2e-8	5.4e-7	8.6e-8	rej.	rej.	rej.
$\alpha = 90°$	2.6e-11	2.0e-9	4.6e-8	rej.	6.6e-8	2.2e-6	rej.	rej.
PH (unsymmetric, hybrid space)								
$\alpha = 0°$	2.2e-14	1.8e-14	3.3e-14	3.2e-14	5.8e-14	1.2e-13	2.8e-13	4.3e-13
$\alpha = 15°$	1.2e-14	4.6e-14	2.0e-14	2.5e-14	1.0e-13	9.8e-14	2.7e-13	3.3e-13
$\alpha = 30°$	6.5e-15	1.0e-14	4.2e-14	4.8e-14	2.1e-13	2.7e-13	3.0e-13	5.1e-13
$\alpha = 45°$	1.3e-14	3.2e-14	3.5e-14	1.2e-13	1.7e-12	2.4e-12	7.8e-12	9.9e-10
$\alpha = 60°$	7.9e-14	1.3e-12	6.3e-13	5.4e-12	2.1e-12	1.4e-10	rej.	rej.
$\alpha = 90°$	1.7e-11	3.3e-11	5.1e-11	rej.	2.9e-11	8.9e-11	rej.	rej.

$\alpha = 60°$, $N = 2$, $p = 8$, with $\sigma_{\min}/\sigma_{\max}(\boldsymbol{K}_{\mathsf{PM}}) = 1.1 \cdot 10^{-15}$ against $4.8 \cdot 10^{-8}$ for PH and $2.9 \cdot 10^{-6}$ for PP; for PH the residual is $\leq 1.2 \cdot 10^{-10}$ and $e_n \leq 8.6 \cdot 10^{-10}$ in all ten cases that are not rejected. For $p < 5$ the polynomial field behaves like a non-polynomial one (rates $2, 3, 4$ for $\alpha \leq 15°$, Fig. 7).

4.3 Non-polynomial fields: convergence

Figures 2 and 3 show the nodal error, Fig. 4 the energy error for the meshes without folded elements; the rates are collected in Tables 4 to 7.

1. $\alpha = 0°$ (element maps asymptotically affine): the three elements converge optimally, $e_E = O(h^p)$ (smooth field: 2.00, 2.99, 4.00, 4.99, 6.00 for PM and PH; Lamé: 1.96–6.69, approaching p from below as the r^{-2} term is resolved), with almost identical errors.

2. $\alpha = 15°$: PM keeps the optimal rates for $p \leq 6$ (smooth: 2.00, 2.99, 3.99, 4.99, 5.98; Lamé: 1.97, 2.91, 3.90, 4.81, 5.83), and its energy error exceeds that on the undistorted mesh by 5–12 % (smooth) and 5–28 % (Lamé); for $p \geq 7$ it reaches a round-off floor of about 10^{-10}, two to three orders above the floor on the undistorted mesh. PH is optimal for all p (smooth: 2.00, 3.00, 4.00, 4.99, 6.01; Lamé: 1.96, 2.94, 3.94, 4.92, 5.91, 6.88, 7.87) and has no elevated floor ($e_E = 6.7 \cdot 10^{-13}$ at $N = 8$, $p = 8$, smooth). PP loses half of the order: its energy rates are 1.08, 1.23, 2.21, 2.03, 2.94, 3.39, 4.05 (smooth) and, for the polynomial field, 0.98, 0.95, 1.95, 1.94, 2.94, 2.95, 3.95, i.e. $\lfloor p/2 \rfloor$. This is the approximation property of the parametric space on maps that are not asymptotically affine: the mid-side node offset is $O(h)$, so $\boldsymbol{x}(\boldsymbol{\xi})$ is quadratic with $O(1)$ relative non-affinity, and $Q_p \circ \boldsymbol{x}^{-1}$ contains $P_k(\boldsymbol{x})$ only for $2k \leq p$ (cf. [1, 2]). At $N = 8$ the PP energy error is 5–$2.6 \cdot 10^3$ times larger than that of PH (smooth, $p = 2, \ldots, 8$).

3. $\alpha = 30°$: PM is optimal only for $p \leq 3$; for $p \geq 4$ its convergence is erratic and even divergent (smooth, $p = 4$: $e_E = 2.2 \cdot 10^{-4}$ at $N = 4$ and $4.3 \cdot 10^{-4}$ at $N = 8$), and at $N = 8$, $p = 8$ it is less accurate than PP ($e_E = 7.3 \cdot 10^{-8}$ against $3.5 \cdot 10^{-9}$). PH remains optimal for all p (smooth: 2.00, 3.00, 4.01, 5.00, 6.02; Lamé: 1.96–7.87); at $N = 8$ its energy error is 8–$2.6 \cdot 10^4$ times smaller than that of PP; compared with PM it is equal at $p = 2$ and 1.4–$6.7 \cdot 10^4$ times smaller for $p \geq 3$ (smooth). PP converges with $\approx \lfloor p/2 \rfloor$.

4. Folded meshes ($\alpha \geq 45°$; for $\alpha = 45°$ from $N = 2$ on): PM converges erratically in the nodal error for all p (Table 6); at $\alpha = 45°$, $N = 8$, $p = 8$ its error is $e_E = 1.6 \cdot 10^{-3}$. PH converges at $\alpha = 45°$ for all p (nodal rates 2.1–7.3, smooth field: 2.55, 3.66, 4.75, 5.55; $e_E = 2.5 \cdot 10^{-9}$ at $N = 8$, $p = 8$) and at $\alpha = 60°$, $90°$ for $p \geq 4$ and $p \geq 5$, respectively. For $p = 2$ at $60°$ and $p \leq 4$ at $90°$ its error is non-monotone in N and of order 10^{-2}–1 (smooth, $\alpha = 60°$, $p = 2$: $e_n = 0.34$ at $N = 8$; Lamé, $\alpha = 90°$, $p = 4$: $e_n = 3.9$ at $N = 2$), as are the errors of the other two elements. PP converges monotonically at the reduced rate for $\alpha = 45°$; on the folded meshes $\alpha \geq 60°$ its convergence is irregular for $p \leq 3$, but its nodal error still decreases by 1.5–3 orders of magnitude between $N = 1$ and $N = 8$ for $p \geq 4$ (Fig. 2).

Under p-refinement on a fixed mesh (Fig. 6), PM converges exponentially for $\alpha \leq 15°$ down to its round-off floor (reached at $p \approx 6$ for $N = 8$), like PP on the undistorted mesh; at $\alpha = 30°$ and $45°$ its error is non-monotone in p (smooth field, $N = 8$, $\alpha = 30°$: e_n jumps from $2.4 \cdot 10^{-6}$ at $p = 3$ to $2.8 \cdot 10^{-5}$ at $p = 4$ and grows from

$5.2 \cdot 10^{-11}$ at $p = 6$ to $4.1 \cdot 10^{-9}$ at $p = 8$). PH converges exponentially down to $3.5 \cdot 10^{-13}$ ($\alpha = 30°$, $p = 7$) and $1.1 \cdot 10^{-11}$ ($\alpha = 45°$, $p = 7$), PP monotonically, but more slowly than on the undistorted mesh.

4.4 Approximation versus stability

The exact nodal values inserted into the trial functions of each element give the interpolation error of its trial space (`run_interpolation_study.m`, `results/raw/interp_a*.csv`); the ratio of discretisation to interpolation error separates the stability of the discrete problem from the approximation property of the space (Table 8). For PH the ratio is 0.78–1.00 in all 158 cases on the meshes without folds in which the interpolation error is above the round-off level (all N, p, both non-polynomial fields), for PP 0.84–1.00: both are quasi-optimal with a constant close to one. For PM it is 0.98–1.01 at $\alpha = 15°$, $p \leq 6$, but ranges from 0.07 to 10^5 overall (0.2–64 at $\alpha = 30°$, $N = 8$); the degraded PM results are therefore not explained by the approximation property of its space. The interpolation errors of the two unsymmetric spaces differ as well. For the Lamé field at $\alpha = 15°$ the PH space approximates worse (interpolation error 1.15–5.7 times that of PM, $N = 8$), which accounts for the larger PH errors there (up to a factor 4.1 in e_E for $p \leq 7$); for the smooth field at $\alpha = 30°$ the PM interpolant is up to 9 times less accurate for $p \leq 6$ because of its large Lebesgue constants (Section 4.6).

4.5 Distortion sensitivity

Table 9 gives the error amplification $e_n(\alpha)/e_n(0°)$ on the finest mesh, Fig. 5 the nodal error versus α. For a non-polynomial field none of the elements is distortion immune: the errors depend on α. For $\alpha = 15°$ and $p \leq 6$ the amplification is 1.5–2.6 (smooth) and 3.2–4.7 (Lamé) for PM, 1.6–2.4 and 6.4–93 for PH, against 4.8–99 and 61–2900 for PP. At 30° the PM amplification reaches $2.2 \cdot 10^3$ (smooth, $p = 4$) and 10^4 (Lamé, $p = 8$), whereas that of PH stays ≤ 4.7 (smooth) and ≤ 381 (Lamé), that of PP reaches $3.4 \cdot 10^3$ and $3.6 \cdot 10^4$. On the folded meshes the PM amplification exceeds the PP amplification by up to six orders of magnitude (Lamé, 45°, $p = 8$); that of PH is smaller than that of PP for all p at 45° and for all $p \geq 3$ at 60° (rejected cases excepted). (For $p \geq 7$ the reference errors at $\alpha = 0°$ are at round-off level, so these columns underestimate the sensitivity.) The amplification of PH for the Lamé field grows with p (7.1 at $p = 2$, 93 at $p = 6$, $\alpha = 15°$): its error constant depends on the distortion, its rate does not (Table 5).

4.6 Mechanism

The diagnostics in `results/` identify where the loss of PM originates and why PH avoids it.

1. *Not the Jacobian, not quadrature.* The loss occurs on meshes with $\det \boldsymbol{J} > 0$ ($\alpha = 30°$) and persists with exact integration (Sections 4.2 and 4.7).

2. *Not consistency, not approximation.* The residual of the exact nodal values, $\boldsymbol{r} = \boldsymbol{K}\boldsymbol{u}_I - \boldsymbol{F}$, is smaller for PM than for PP in all eleven cases tested (`residual_smooth.txt`; e.g. $\alpha = 30°$, $N = 8$, $p = 4$: $3.0 \cdot 10^{-5}$ versus $3.7 \cdot 10^{-4}$), whereas the PM nodal error is 9 times larger ($2.8 \cdot 10^{-5}$ versus $3.0 \cdot 10^{-6}$); PH has the residual $3.3 \cdot 10^{-6}$ and the nodal error $5.1 \cdot 10^{-8}$. The ratio of discretisation to interpolation error of PM is far from one where its results degrade (Section 4.4).

3. *Stability of the unsymmetric operator.* With $\boldsymbol{u}_h - \boldsymbol{u}_I = -\boldsymbol{K}^{-1}\boldsymbol{r} = -\sum_k \boldsymbol{v}_k(\boldsymbol{u}_k^{\mathsf{T}}\boldsymbol{r})/\sigma_k$ (singular triplets of $\boldsymbol{K}$; `spectrum_smooth.txt`, smooth field), the error of PP, of PH (all cases) and of PM for $\alpha \leq 15°$ is spread over the spectrum (largest single contribution 2–8 % of $\|\boldsymbol{u}_h - \boldsymbol{u}_I\|^2$, from mid-spectrum modes). For $\alpha \geq 30°$, PM has singular values below the smallest singular value of PP (down to $0.17\,\sigma_1^{\mathsf{PP}}$, $\alpha = 45°$, $N = 2$, $p = 4$), and in four of the five cases tested a single one of the lowest six triplets carries 21–68 % of the error (the fifth, $\alpha = 30°$, $N = 4$, $p = 6$: triplet $k = 28$, 9 %). The residual of the Petrov–Galerkin scheme is not orthogonal to these low modes, so they are excited and amplified. The lowest singular values of PH agree with those of PP within 2.3 % ($k = 1, \ldots, 50$) in all seven cases. At the upper end of the spectrum the difference is larger (`smin.txt`, symmetry conditions of the Lamé problem, 14 cases with $\alpha \leq 30°$): the smallest singular values of the three global matrices agree to 1 %, but $\sigma_{\max}(\boldsymbol{K}_{\mathsf{PM}})$ exceeds $\sigma_{\max}(\boldsymbol{K}_{\mathsf{PP}})$ by factors up to 20 ($\alpha = 30°$, $N = 8$, $p = 4$) and 344 ($\alpha = 30°$, $N = 4$, $p = 6$), whereas $\sigma_{\max}(\boldsymbol{K}_{\mathsf{PH}})/\sigma_{\max}(\boldsymbol{K}_{\mathsf{PP}}) \leq 1.2$.

4. *Ill-conditioned metric interpolation.* The Lebesgue constant $\Lambda_M = \max_{\boldsymbol{x} \in \Omega^e} \sum_i |M_i^e(\boldsymbol{x})|$ of the trial functions (basis independent; sampled at the images of a 16×16 grid of the reference element, so the values are lower bounds) equals the GLL value on affine elements, but grows by orders of magnitude for PM with α and p (Table 10): $\Lambda_M \leq 130$ for $\alpha = 15°$, $p \leq 6$, but $1.1 \cdot 10^3$ ($\alpha = 30°$, $p = 4$), $5 \cdot 10^5$ (45°, $p = 5$) and $1.3 \cdot 10^8$ (45°, $p = 8$). The element condition numbers of PM exceed those of PP by

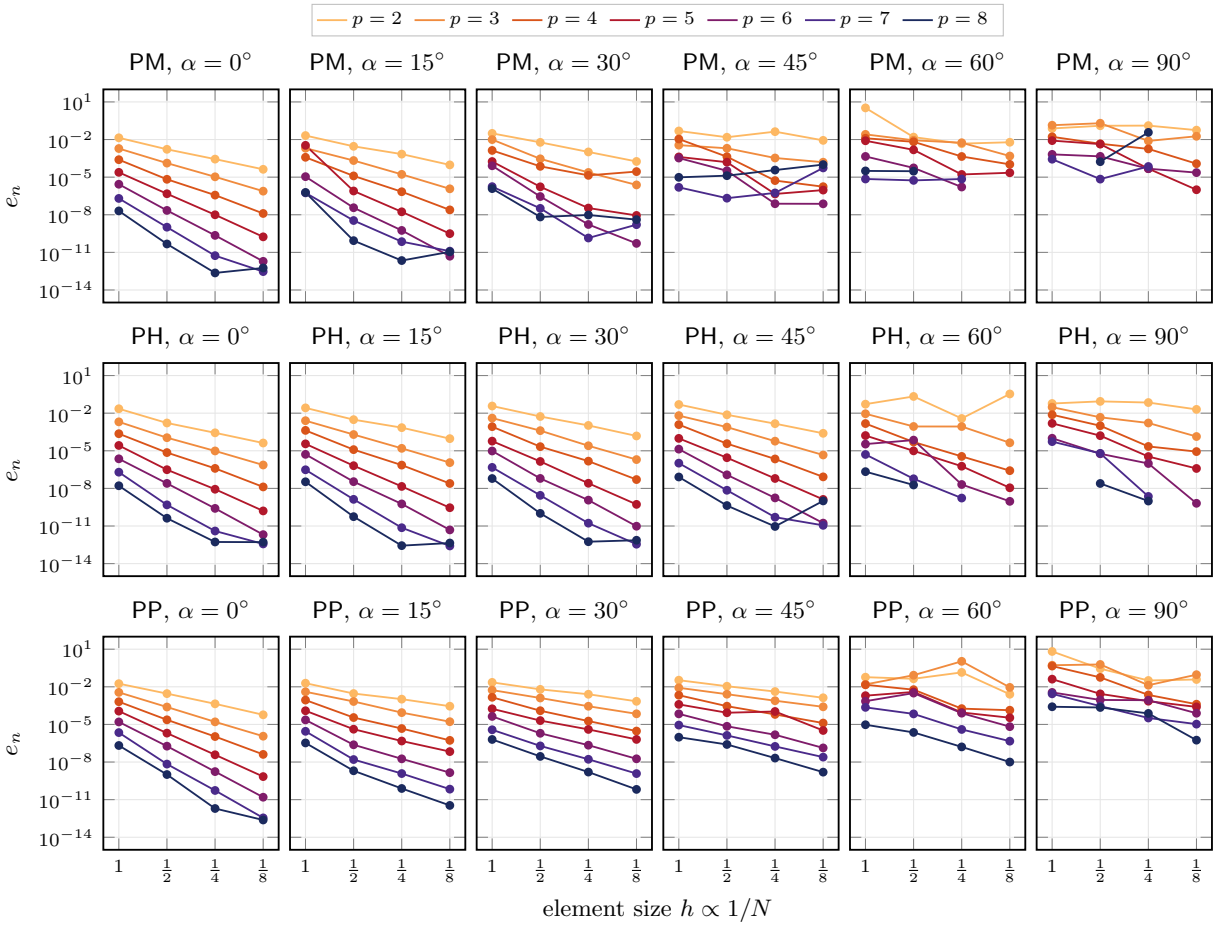


Figure 2: Smooth manufactured field: nodal error e_n versus element size for all distortion angles, PM (top), PH (middle) and PP (bottom).

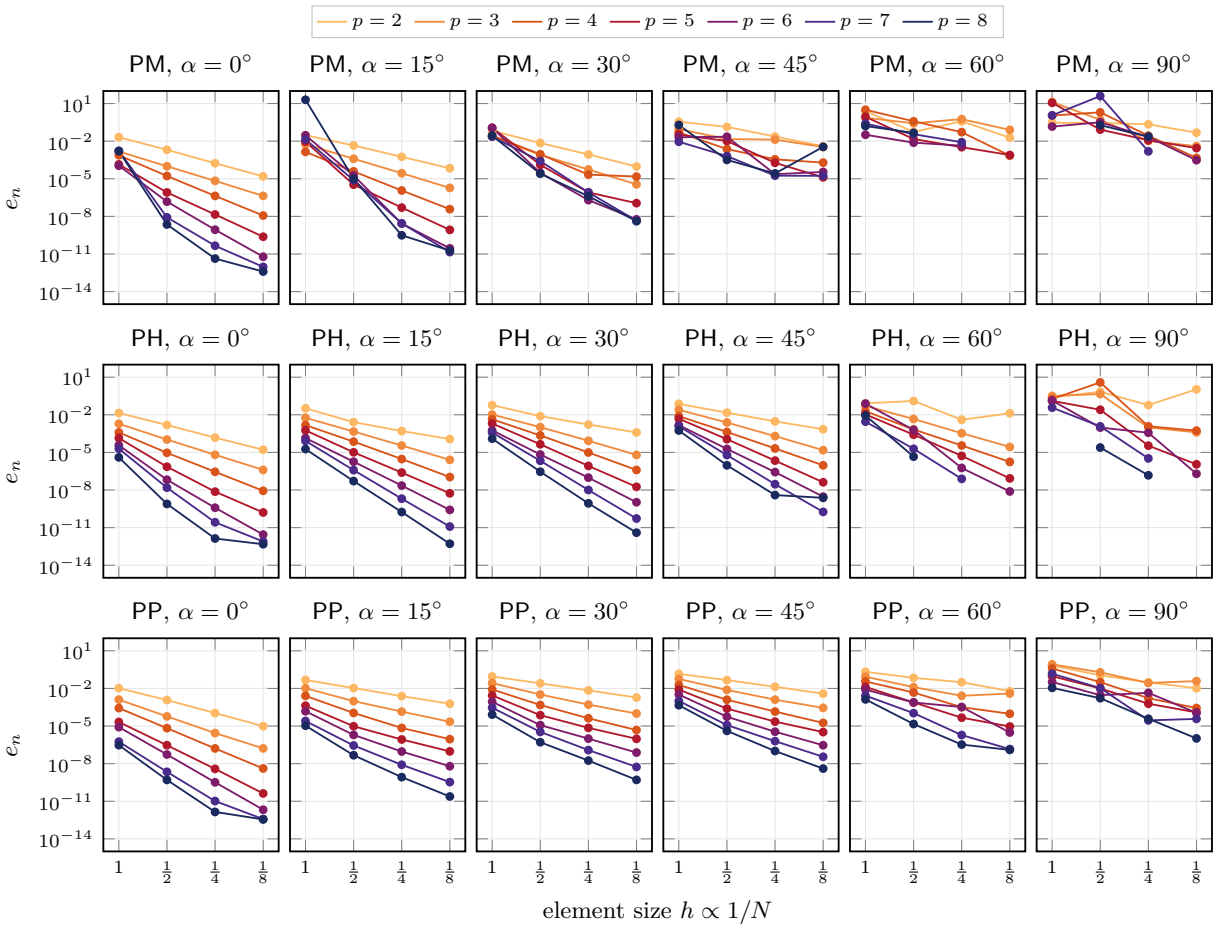


Figure 3: Thick cylinder (Lamé): nodal error e_n versus element size, PM (top), PH (middle) and PP (bottom).

Table 4: Rates of e_E, smooth field ($N = 4 \to 8$).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 0°$	2.00	2.99	4.00	4.99	6.00	–	–
$\alpha = 15°$	2.00	2.99	3.99	4.99	5.98	–	–
$\alpha = 30°$	2.00	2.86	-0.98	2.40	–	–	–
PH (unsymmetric, hybrid space)							
$\alpha = 0°$	2.00	2.99	4.00	4.99	6.00	–	–
$\alpha = 15°$	2.00	3.00	4.00	4.99	6.01	–	–
$\alpha = 30°$	2.00	3.00	4.01	5.00	6.02	–	–
PP (conventional)							
$\alpha = 0°$	2.02	3.19	4.22	5.37	6.39	7.43	–
$\alpha = 15°$	1.08	1.23	2.21	2.03	2.94	3.39	4.05
$\alpha = 30°$	1.02	1.06	2.01	1.98	2.98	3.11	3.94

Table 5: Rates of e_E, Lamé ($N = 4 \to 8$).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 0°$	1.97	2.90	3.89	4.78	5.77	6.62	–
$\alpha = 15°$	1.97	2.91	3.90	4.81	5.83	–	–
$\alpha = 30°$	1.97	2.94	0.76	2.99	4.56	–	–
PH (unsymmetric, hybrid space)							
$\alpha = 0°$	1.96	2.91	3.89	4.81	5.79	6.69	–
$\alpha = 15°$	1.96	2.94	3.94	4.92	5.91	6.88	7.87
$\alpha = 30°$	1.96	2.96	3.94	4.94	5.92	6.90	7.87
PP (conventional)							
$\alpha = 0°$	1.91	2.89	3.85	4.79	5.75	6.67	7.60
$\alpha = 15°$	1.07	1.43	2.16	2.81	3.49	4.19	4.78
$\alpha = 30°$	1.03	1.21	2.07	2.51	3.20	3.85	4.42

Table 6: Rates of e_n, smooth field ($N = 4 \to 8$).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 0°$	2.70	3.80	4.86	5.84	6.88	–	–
$\alpha = 15°$	2.86	3.83	4.81	5.79	6.83	–	–
$\alpha = 30°$	2.52	3.30	-1.00	1.96	–	–	–
$\alpha = 45°^\dagger$	2.31	1.12	1.62	-1.02	–	–	–
$\alpha = 60°^\dagger$	-0.37	3.52	2.08	-0.45	–	–	–
$\alpha = 90°^\dagger$	1.20	-1.23	3.92	5.59	–	–	–
PH (unsymmetric, hybrid space)							
$\alpha = 0°$	2.70	3.72	4.91	5.75	6.95	–	–
$\alpha = 15°$	2.91	3.78	4.87	5.64	6.88	–	–
$\alpha = 30°$	2.79	3.66	4.88	5.60	6.91	–	–
$\alpha = 45°^\dagger$	2.55	3.66	4.75	5.55	–	–	–
$\alpha = 60°^\dagger$	-6.50	4.30	3.78	5.71	–	–	–
$\alpha = 90°^\dagger$	1.81	3.59	1.36	3.16	–	–	–
PP (conventional)							
$\alpha = 0°$	2.91	3.80	4.73	5.78	6.79	–	–
$\alpha = 15°$	1.87	2.41	3.14	2.80	3.66	4.13	4.49
$\alpha = 30°$	1.87	2.00	2.65	2.65	3.61	3.79	4.59
$\alpha = 45°^\dagger$	1.65	1.62	2.28	5.03	3.49	2.84	3.73
$\alpha = 60°^\dagger$	5.79	6.91	0.42	1.39	3.57	3.10	4.01
$\alpha = 90°^\dagger$	-0.26	-2.79	2.45	1.51	3.46	1.62	7.08

Table 7: Rates of e_n, Lamé ($N = 4 \to 8$).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 0°$	3.48	3.98	5.21	5.90	7.15	–	–
$\alpha = 15°$	3.03	3.90	4.96	5.90	6.57	–	–
$\alpha = 30°$	3.19	3.92	0.51	2.85	5.09	–	–
$\alpha = 45°^\dagger$	2.57	1.95	0.87	3.85	-0.59	–	–
$\alpha = 60°^\dagger$	4.35	2.84	6.22	2.08	–	–	–
$\alpha = 90°^\dagger$	2.23	1.42	5.90	2.08	6.16	–	–
PH (unsymmetric, hybrid space)							
$\alpha = 0°$	3.19	4.01	5.02	5.52	7.12	–	–
$\alpha = 15°$	2.12	3.79	4.78	5.50	6.45	7.39	–
$\alpha = 30°$	2.10	3.75	4.64	5.54	6.52	7.56	–
$\alpha = 45°^\dagger$	2.09	3.72	4.48	5.74	6.52	7.34	–
$\alpha = 60°^\dagger$	-1.73	3.63	4.35	5.96	6.24	–	–
$\alpha = 90°^\dagger$	-4.20	1.30	1.15	4.94	10.87	–	–
PP (conventional)							
$\alpha = 0°$	3.47	4.06	5.31	6.50	7.27	–	–
$\alpha = 15°$	2.02	2.70	2.96	3.15	3.90	4.58	5.13
$\alpha = 30°$	1.88	2.36	3.12	2.91	3.69	4.46	5.12
$\alpha = 45°^\dagger$	1.81	2.17	3.00	2.78	3.59	4.14	4.65
$\alpha = 60°^\dagger$	2.37	-0.59	1.74	2.35	6.75	3.72	1.40
$\alpha = 90°^\dagger$	1.55	-0.48	2.62	2.30	5.29	-0.39	5.21

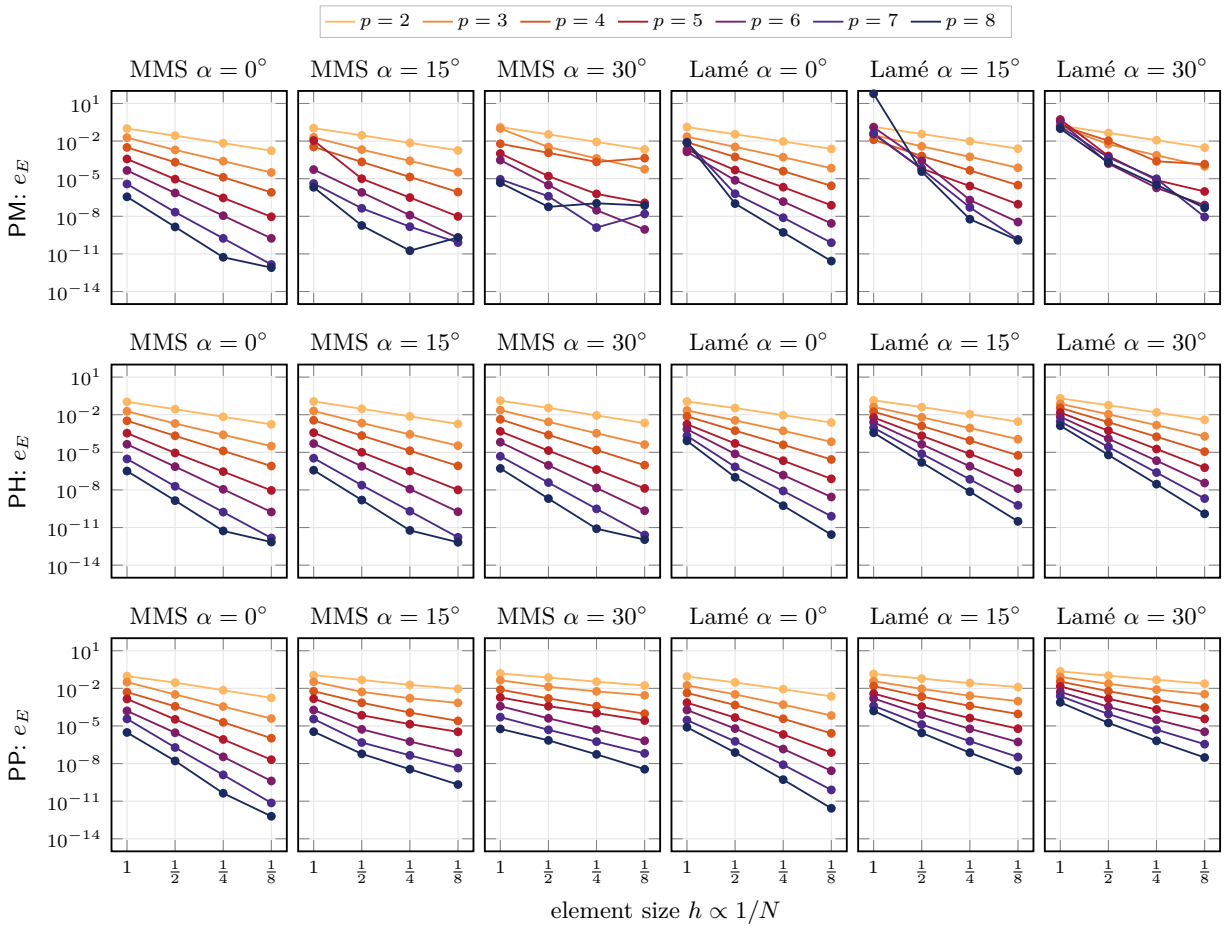


Figure 4: Energy error e_E on the meshes without folded elements ($\alpha \leq 30°$): smooth field (left three columns) and Lamé (right three columns); PM top, PH middle, PP bottom.

similar factors (Table 11). The images of the GLL nodes of strongly non-affine elements come close to configurations that are not unisolvent for Q_p in skew coordinates ("interpolation failure" in [5]); the metric trial functions then have large gradients, the bilinear form loses its stability constant, and round-off is amplified (Section 4.2). All cases of the benign regime of Section 4.3 have $\Lambda_M \leq 2.4 \cdot 10^2$, all degraded cases $\Lambda_M \geq 10^3$; the converse does not hold strictly ($\alpha = 30°$, $p = 6$: $\Lambda_M = 6.2 \cdot 10^3$, yet PM is 730 times more accurate than PP at $N = 8$, smooth), so Λ_M is a necessary rather than a sufficient indicator.

5. *Hybrid space.* PH uses metric functions only for the complete polynomials $P_p(\boldsymbol{x})$, the remaining functions are parametric. Its Lebesgue constant is the GLL value up to 15° (≤ 4.5), ≤ 95 at 30° and ≤ 546 at 45° (Table 10), its element condition numbers are those of PP within a factor 1.8 for $\alpha \leq 30°$, $p \geq 3$ (Table 11), and the conditioning of its interpolation matrix is much better than that of PM at 30°, $p \geq 6$ (factors 10^2–10^5; $\kappa(\boldsymbol{V}) = 2.7 \cdot 10^3$ against $2.6 \cdot 10^8$ at $p = 7$). The large Lebesgue constants of PM are therefore caused by the high-degree terms of $Q_p(\hat{x}, \hat{y})$ beyond P_p, which are not needed for completeness. On folded elements (Table 10, $\alpha = 90°$: $\Lambda \leq 4.8 \cdot 10^3$) and for $p = 2$ at $\alpha \geq 30°$ (Table 11: 23–117) also PH is degraded, and for some folded elements with $p \geq 7$ its interpolation matrix is singular (Section 4.2): there the nodes are not unisolvent for T_p^e either.

4.7 Quadrature

The integrand (1) of PM is a polynomial of degree $(2p-1)(q+1)$ per direction for a geometry of degree q; exact integration needs $n_q \geq p^2 + p/2$ points on the elements with exact arcs ($q = p$) and $n_q \geq 3p - 1$ on the interior elements ($q = 2$). The complement of the PH trial functions involves $\boldsymbol{J}^{-1}$, so its integrand is rational and is not integrated exactly by any rule. The default rule ($n_q = 2p$) is therefore not exact. Repeating selected cases with $n_q = 2p, 3p, 4p, 6p$ (`results/raw/quadrature.csv`; $\alpha = 0°, 30°$, $N = 1, 2$, $p = 5, 8$) changes e_E of the non-polynomial fields by at most $3 \cdot 10^{-7}$ (relative) for PP, $5 \cdot 10^{-6}$ for PH and 0.5 % for PM; the single

Table 8: Discretisation error over interpolation error in the same trial space, e_E, $N = 8$: smooth field (left) and Lamé (right).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 15°$	0.99	0.98	1.00	0.98	1.01	–	–
$\alpha = 30°$	0.91	0.20	64.38	3.40	0.38	–	–
PH (unsymmetric, hybrid space)							
$\alpha = 15°$	0.99	0.97	0.98	0.96	0.98	–	–
$\alpha = 30°$	0.96	0.93	0.93	0.89	0.90	–	–
PP (conventional)							
$\alpha = 15°$	0.97	0.94	0.96	0.93	0.96	0.95	0.96
$\alpha = 30°$	0.94	0.92	0.92	0.89	0.93	0.93	0.93

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 15°$	1.00	0.99	0.99	0.98	0.99	–	–
$\alpha = 30°$	0.99	0.48	14.98	5.46	8.75	–	–
PH (unsymmetric, hybrid space)							
$\alpha = 15°$	0.98	0.95	0.96	0.95	0.96	0.94	0.95
$\alpha = 30°$	0.93	0.91	0.92	0.89	0.90	0.88	0.89
PP (conventional)							
$\alpha = 15°$	0.98	0.96	0.98	0.97	0.98	0.97	0.98
$\alpha = 30°$	0.95	0.93	0.96	0.93	0.95	0.94	0.95

Table 9: Distortion sensitivity on the finest mesh ($N = 8$): $e_n(\alpha)/e_n(0°)$, smooth field (left) and Lamé (right).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 15°$	2.25	1.54	1.93	1.8	2.56	42.8	19.3
$\alpha = 30°$	4.24	3.15	2.2e3	50.9	27	5.5e3	7.1e3
$\alpha = 45°^\dagger$	204	204	137	5.4e3	3.9e4	1.9e8	1.7e8
$\alpha = 60°^\dagger$	144	636	8.3e3	1.3e5	rej.	rej.	rej.
$\alpha = 90°^\dagger$	1.3e3	2.4e4	9.4e3	5.7e3	1.2e7	rej.	rej.
PH (unsymmetric, hybrid space)							
$\alpha = 15°$	2.23	1.55	1.88	1.8	2.39	0.689	0.855
$\alpha = 30°$	3.66	2.71	3.87	3.32	4.7	0.949	1.42
$\alpha = 45°^\dagger$	5.94	6.3	6.34	8.28	8.3	30.3	1.9e3
$\alpha = 60°^\dagger$	8.2e3	59.4	19.8	70	452	rej.	rej.
$\alpha = 90°^\dagger$	480	187	659	2.4e3	315	rej.	rej.
PP (conventional)							
$\alpha = 15°$	4.84	14.4	13.2	98.7	91.1	196	14.5
$\alpha = 30°$	11.9	61.2	72.8	923	1.2e3	3.4e3	276
$\alpha = 45°^\dagger$	22.8	221	317	4.8e3	8.6e3	7.0e4	6.6e3
$\alpha = 60°^\dagger$	43.5	7.8e3	3.3e3	4.9e4	4.3e5	1.3e6	4.2e4
$\alpha = 90°^\dagger$	654	8.1e4	1.0e4	3.6e5	5.0e6	3.0e7	2.3e6

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 15°$	4.45	4.31	3.22	3.54	4.65	15.2	47.3
$\alpha = 30°$	6.11	8.3	1.3e3	479	959	4.8e3	1.0e4
$\alpha = 45°^\dagger$	253	7.9e3	1.7e4	5.5e4	5.8e6	1.8e7	8.8e9
$\alpha = 60°^\dagger$	1.2e3	1.8e5	6.0e4	3.3e6	rej.	rej.	rej.
$\alpha = 90°^\dagger$	3.0e3	9.1e3	3.9e4	1.2e7	5.0e7	rej.	rej.
PH (unsymmetric, hybrid space)							
$\alpha = 15°$	7.1	6.35	12.4	33.8	93.3	15.5	1.09
$\alpha = 30°$	23.9	15.6	45.6	111	381	67.9	8.32
$\alpha = 45°^\dagger$	42.9	37.5	107	256	1.1e3	230	5.0e3
$\alpha = 60°^\dagger$	811	68	201	524	2.8e3	rej.	rej.
$\alpha = 90°^\dagger$	6.6e4	991	6.3e4	6.9e3	7.1e4	rej.	rej.
PP (conventional)							
$\alpha = 15°$	61.3	132	218	2.3e3	2.9e3	888	67.5
$\alpha = 30°$	194	601	1.2e3	2.3e4	3.6e4	1.4e4	1.4e3
$\alpha = 45°^\dagger$	392	1.7e3	4.3e3	7.8e4	1.4e5	9.3e4	1.1e4
$\alpha = 60°^\dagger$	619	2.4e4	2.3e4	2.2e5	1.4e6	3.8e5	3.5e5
$\alpha = 90°^\dagger$	1.1e3	2.3e5	6.6e4	2.9e6	5.4e7	9.7e7	2.9e6

exception (PM, smooth field, $\alpha = 30°$, $N = 2$, $p = 8$: e_E varies by a factor 2.7) has an error of the order of the round-off level of that mesh (polynomial field with the same rules: $6 \cdot 10^{-9}$–$7 \cdot 10^{-8}$). For the polynomial field ($p \geq 5$), whose error is pure round-off, the value changes by up to a factor 21 for PM and 2.7 for PH (at $\leq 10^{-13}$). The conclusions are therefore not affected by quadrature. On folded meshes the PM results change erratically with n_q (e.g. Lamé, $\alpha = 90°$, $N = 1$, $p = 5$: $e_E = 38$ for $n_q = 10$, 0.66 for $n_q \geq 15$), consistent with the round-off amplification of Section 4.2.

5 Answers to the key questions

Is distortion resistance lost for non-polynomial fields? Distortion *immunity* in the sense of [5] (results independent of the mesh) is lost by construction: the error of a non-polynomial field depends on the mesh, for both fields studied and for both unsymmetric elements (Table 9). For PM, distortion *resistance* depends on the distortion level and on p:

- moderate distortion without folds ($\alpha = 15°$, $p \leq 6$; $\alpha = 30°$, $p \leq 3$): retained. The energy error of PM exceeds the error on the undistorted mesh by 5–28 % (15°) and 26–82 % (30°, $p \leq 3$), whereas the nodal error of the conventional element is 3–290 times larger ($N = 8$);
- strong distortion ($\alpha = 30°$ with $p \geq 4$, $\alpha \geq 45°$) or $p \geq 7$: lost. The PM error becomes erratic and in several cases larger than that of the conventional element (e.g. $\alpha = 30°$, $N = 8$, $p = 4$ and 8; $\alpha = 45°$, $N = 8$, $p = 7, 8$); on folded meshes its convergence is erratic for all p (Table 6), and for $p \geq 7$ at $\alpha \geq 15°$ it stalls at an elevated round-off floor.

The loss is not tied to the sign of $\det \boldsymbol{J}$, the quantity the formulation was designed to be insensitive to: it already occurs on meshes with $\det \boldsymbol{J} > 0$ ($\alpha = 30°$) and correlates with the Lebesgue constant of the metric interpolation (Table 10).

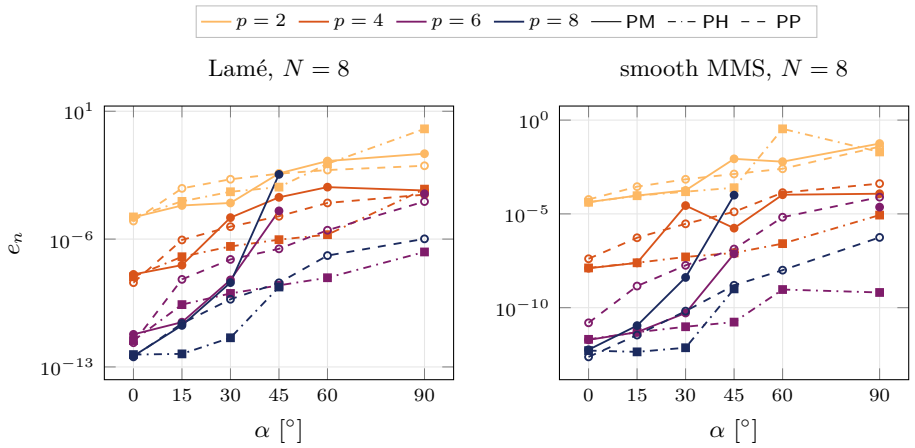


Figure 5: Nodal error versus distortion angle on the finest mesh ($N = 8$), $p = 2, 4, 6, 8$; PM solid, PH dash-dotted, PP dashed. Folded elements occur for $\alpha \geq 45°$.

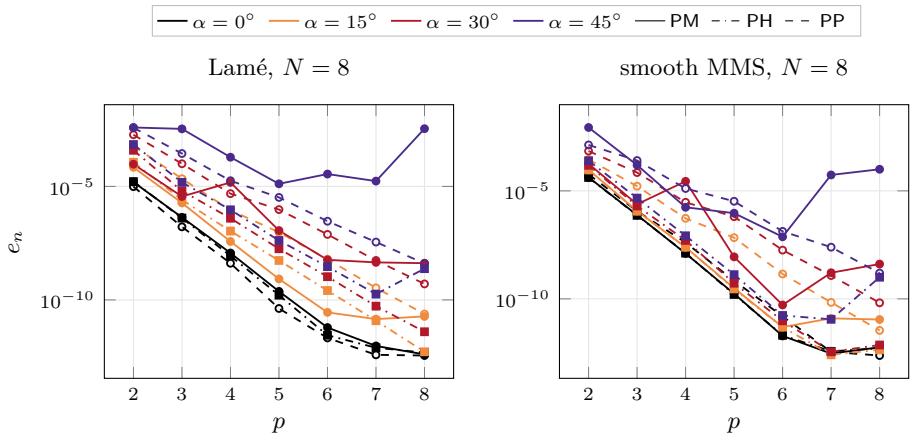


Figure 6: p-convergence of the nodal error on the finest mesh ($N = 8$), $\alpha = 0°, 15°, 30°, 45°$; PM solid, PH dash-dotted, PP dashed.

Are optimal convergence rates retained? For PM in the moderate regime, yes: it converges with the optimal energy-norm order p for $p = 2, \ldots, 6$ at $\alpha = 15°$ (and $p = 2, 3$ at $30°$) for the analytic manufactured field and for the Lamé field, and exponentially under p-refinement down to a round-off floor of about 10^{-10}. The conventional element loses half of the order on the same meshes ($e_E = O(h^{\lfloor p/2 \rfloor})$), because the element maps are not asymptotically affine. In the strong regime, optimal convergence of PM is lost as well: rates are erratic and the error can grow under h- and p-refinement.

Practical benefit of PM. Where both properties hold (moderate, non-folding distortion that persists under refinement), the benefit over PP is substantial and grows with p: optimal order versus $\lfloor p/2 \rfloor$, i.e. an error ratio $O(h^{-\lceil p/2 \rceil})$ (a factor 380 for $p = 6$, $N = 8$, $\alpha = 15°$, smooth). It is limited in the situations for which the formulation was proposed: on folded meshes PM is not usable at high order; whether a mesh and degree fall into the benign regime cannot be inferred from the Jacobian (non-monotone convergence already at $\alpha = 30°$, $p = 4$); for meshes that become asymptotically affine under refinement PP recovers the optimal order, and the advantage reduces to a constant; and the unsymmetric system requires a non-symmetric solver and a generalised Vandermonde solve per element.

Does the hybrid trial space remove the deficiencies? To a large extent, yes. With metric polynomials only for $P_p(\boldsymbol{x})$ and parametric functions for the remainder, PH

- reproduces the degree-5 field to $\leq 9.9 \cdot 10^{-10}$ on all meshes, folded ones included, where PM is limited to 10^{-4}–10^{-2} by round-off (Section 4.2);
- converges with the optimal order for all $p \leq 8$ on all meshes without folds, and at $\alpha = 45°$ (folded for $N \geq 2$) for all p (Section 4.3);

Table 10: Lebesgue constant of the trial functions of PM and PH, maximum over the elements, $N = 8$ (GLL: affine element; 16×16 sample points).

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 0°$	1.56	2.22	2.68	3.02	3.42	4.6	5.35
$\alpha = 15°$	2.53	5.98	15.5	37.2	130	1.2e4	1.3e4
$\alpha = 30°$	9.95	240	1.1e3	1.6e3	6.2e3	4.2e6	1.2e6
$\alpha = 45°^{\dagger}$	186	1.5e3	1.3e4	5.0e5	8.3e5	1.4e6	1.3e8
$\alpha = 60°^{\dagger}$	290	1.1e4	8.5e4	2.8e5	rej.	rej.	rej.
$\alpha = 90°^{\dagger}$	61.2	3.4e3	8.3e5	2.2e5	3.3e5	rej.	rej.
PH (unsymmetric, hybrid space)							
$\alpha = 0°$	1.58	2.22	2.69	3	3.41	3.47	4.15
$\alpha = 15°$	1.76	2.4	2.8	3.1	3.47	4.47	4.18
$\alpha = 30°$	2	4.77	7.52	16	24.9	74.5	94.6
$\alpha = 45°^{\dagger}$	2.78	10	21.2	46.7	91	258	546
$\alpha = 60°^{\dagger}$	4.26	18.9	53.3	193	833	rej.	rej.
$\alpha = 90°^{\dagger}$	389	2.9e3	4.8e3	1.7e3	2.2e3	rej.	rej.
GLL	1.56	2.22	2.68	3	3.41	3.47	4.15

Table 11: Ratio of the maximum element condition numbers $\sigma_{\max}/\sigma_{2n-3}$ to that of PP, $N = 8$.

	p						
	2	3	4	5	6	7	8
PM (unsymmetric, skew coordinates)							
$\alpha = 0°$	0.998	1.01	0.996	1	0.997	1.01	0.998
$\alpha = 15°$	1.54	0.937	1.19	1.61	3.25	182	128
$\alpha = 30°$	60	11.5	145	623	90.3	1.4e5	2.1e6
$\alpha = 45°^{\dagger}$	44.4	12.8	148	7.2e4	1.1e5	6.0e6	2.3e6
$\alpha = 60°^{\dagger}$	10.4	10.7	104	7.2e3	rej.	rej.	rej.
$\alpha = 90°^{\dagger}$	9.95	198	23.2	5.7e3	5.8e4	rej.	rej.
PH (unsymmetric, hybrid space)							
$\alpha = 0°$	1.01	1	1	1	1	1	1
$\alpha = 15°$	1.31	1.03	1.01	0.992	0.997	1.01	1
$\alpha = 30°$	76.3	1.19	1.19	1.01	0.987	1.7	1.76
$\alpha = 45°^{\dagger}$	70.4	19	1.35	3.22	1.71	3.38	6.9
$\alpha = 60°^{\dagger}$	23.3	1.61	43.6	17.6	89.3	rej.	rej.
$\alpha = 90°^{\dagger}$	117	158	1.25	8.61	17.6	rej.	rej.

- is quasi-optimal with constant 0.78–1.00 relative to its interpolation error (Section 4.4), and has Lebesgue constants, element condition numbers and stiffness spectra close to those of PP (Section 4.6);
- is 8–$2.6 \cdot 10^4$ times more accurate than PP and, for $p \geq 3$, 1.4–$6.7 \cdot 10^4$ times more accurate than PM at $\alpha = 30°$, $N = 8$ (smooth field).

It does not remove all limits. (a) Its error constant depends on the distortion: the amplification $e_n(\alpha)/e_n(0°)$ for the Lamé field reaches 93 at $\alpha = 15°$, $p = 6$, and where PM is stable, PH can be less accurate (Lamé, $\alpha = 15°$, $p \leq 7$: up to a factor 4.1), because the parametric complement approximates worse than the full Q_p in skew coordinates (Section 4.4). (b) On folded meshes its low-order results ($p = 2$ at 60°, $p \leq 4$ at 90°) are as unreliable as those of the other elements. (c) Its interpolation matrix is singular for some folded elements at $p \geq 7$, and its complement requires $\det \boldsymbol{J} \neq 0$ at the quadrature points. Within these limits PH is the better choice than PM on every mesh studied, and than PP on every distorted mesh studied.

6 Limitations

One geometry (thick cylinder), one family of distortions (self-similar rotation of mid-side nodes), linear plane-stress elasticity, $p \leq 8$, $N \leq 8$, double precision. The Lebesgue constants are sampled at 16×16 points per element and are therefore lower bounds; they are basis independent, the round-off level is not (Lagrange basis on the bounding box of the skew coordinates, Section 2.1). The energy norm is evaluated with the signed $\det \boldsymbol{J}$; on folded meshes the nodal error is the only quantity that is defined independently of the element mapping. The mechanism of the stability loss (Section 4.6) is characterised numerically, not proven.

7 Reproduction

The first version of this report was computed with the code base of commit `2fea864` (monomial basis of the metric functions, no hybrid space). The present results were computed with the code base after commit `e293da1`, which includes the singularity checks of `ShapeFunctions2D_Metric` and `pSpectralElement_NodesTopology`, the Lagrange basis of the metric functions and the hybrid trial space (`p.metricMethod = 2`), and a vectorised evaluation of `LagrangePolynomials` (bitwise identical results). Rejected cases are recorded as NaN.

`mesh/gen_cylinder_meshes.py`	gmsh model, meshes, ABAQUS files, plot data
`matlab/run_study.m`	main study (MATLAB or GNU Octave)
`matlab/run_interpolation_study.m`	interpolation errors in the trial spaces
`matlab/run_quadrature_study.m`	quadrature sensitivity
`matlab/cyl_verify.m`	cross-checks against the code base
`matlab/cyl_conditioning_check.m`, `cyl_residual_check.m`, `cyl_spectrum_check.m`, `cyl_smin.m`, `cyl_element_analysis.m`	diagnostics of Section 4.6
`postprocess.py`	tables and figure code

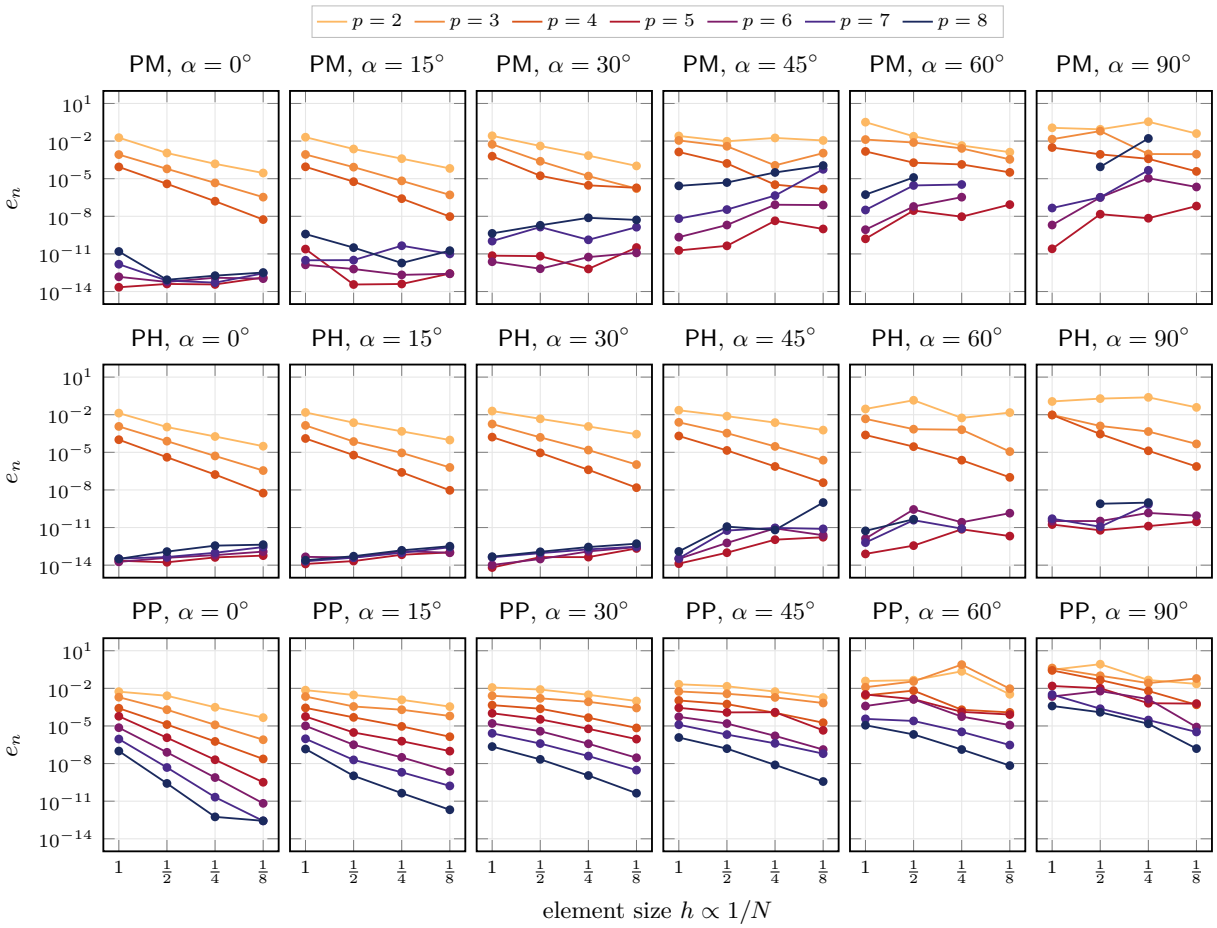


Figure 7: Polynomial field (degree 5): nodal error versus element size, PM (top), PH (middle) and PP (bottom). PM and PH reproduce the field for $p \geq 5$ up to round-off.

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